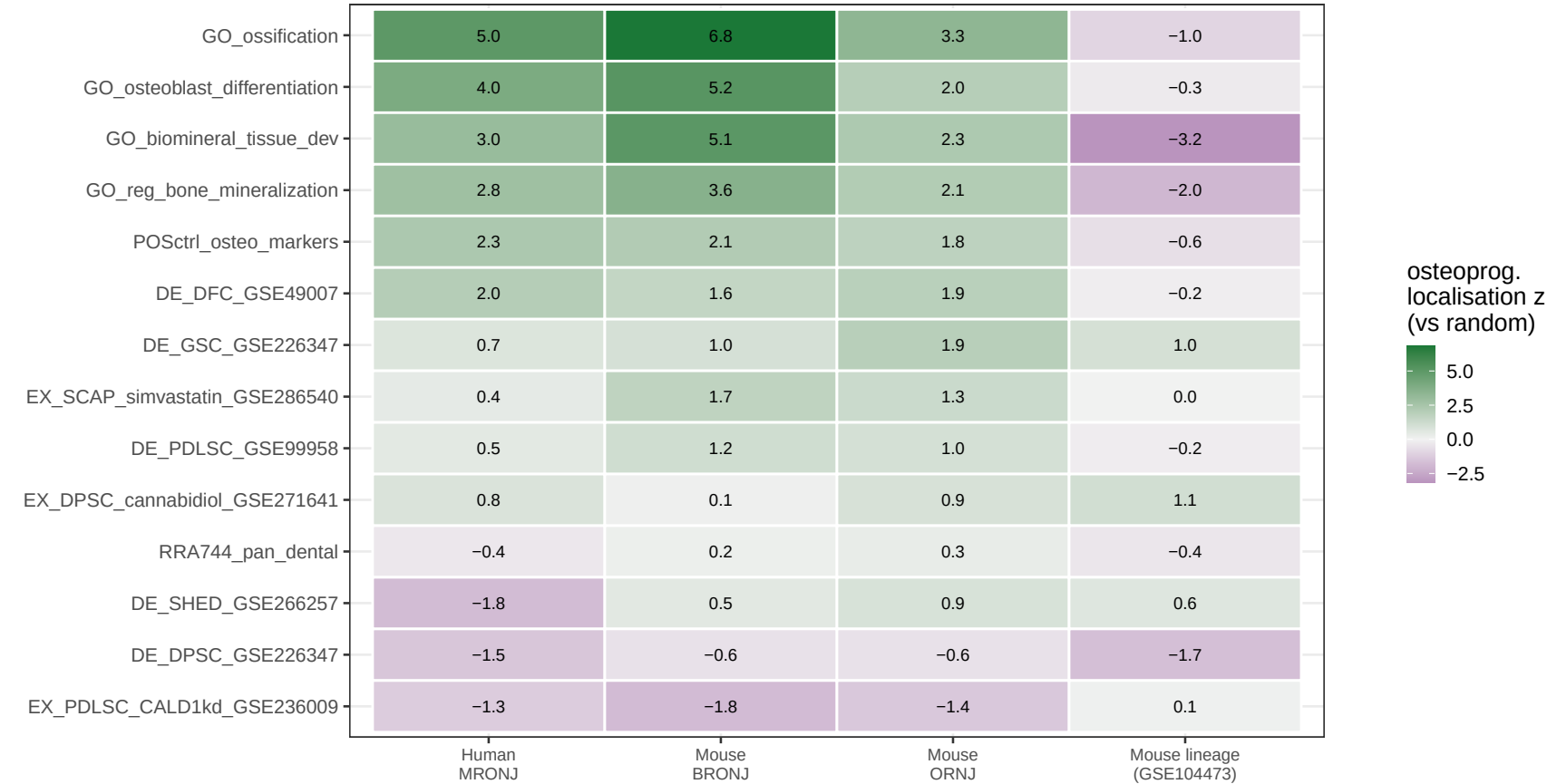


A

Osteoprogenitor localisation of each signature across in-vivo contexts

Calibrated z vs size-matched permutation null (>0 = localises to the osteoprogenitor/mesenchymal compartment more than random). The three scRNA jaw-disease atlases are positive-control-validated; the bulk lineage is not (even canonical markers do not rise SSC->BCSP).



B

Transferability Score (mean osteoprogenitor-localisation z across 3 validated scRNA atlases)

Prior-knowledge osteogenic gene sets and canonical markers localise to in-vivo osteoprogenitors in ALL validated atlases; data-driven in-vitro signatures (incl. the robust RRA-744 meta) do not (transfer by provenance: prior-knowledge 4/4 vs data-driven 1/9, Fisher p=0.007).

